

Face Recognition Implementation Based on Image Processing Techniques

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Certificate

This is to certify that the project titled “**Face Recognition Implementation Based on Image Processing Techniques**” has been carried out by R. Sumanaswi , B. Bikki Bindu Venkata Priya and S. Sana under my supervision in partial fulfillment of the requirements for the Digital Image Processing course. The work has been completed collaboratively by the team.

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Face Recognition Implementation Based on Image Processing Techniques

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Abstract

Facial recognition is one of the important applications of image processing and computer vision and has been used in many areas such as security systems, surveillance cameras, finance industries, and consumer technologies. Although there have been extensive studies on individual recognition algorithms, studies comparing these models using different data sets and different image pre-processing approaches are scarce. In this research, a comparison between three well-known facial recognition models, Principal Component Analysis (PCA), Local Binary Pattern Histograms (LBPH), and FaceNet models, is performed using three benchmark data sets; the AT&T Database of Faces (400 images with 40 subjects), the FEI Face Dataset (2800 images with 200 subjects), and the Faces94 Dataset (3059 images with 152 subjects). These three models were tested using six different pre-processing techniques including non-preprocessing (original), median filtering, grayscale normalization, geometric normalization, wavelet transformation, and histogram equalization. For evaluation purposes, the accuracy, precision, recall, F1-score, Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM) were utilized. Results revealed that the FaceNet model had the highest accuracy rate among all models especially on controlled data sets, where it achieved perfect recognition accuracy on the AT&T data set (100

Keywords: Face recognition, Image processing, PCA, Eigenfaces, LBPH, FaceNet, Deep learning, Preprocessing, benchmark Evaluation, Computer Vision

1. Introduction

The task of face recognition is among the most widely explored biometric technologies in literature and industry use cases. It finds application in border security, financial transactions authentication, smartphone unlocking, patient verification in hospitals, and physical access control in secure facilities [1]. In contrast to password-based and token-based approaches to authentication, face recognition is a non-invasive technique and can be performed passively on large groups of people. This characteristic explains its rapid diffusion.

Overall, face recognition pipeline consists of several main phases: imaging capture, preprocessing, feature extraction, and finally, identity matching. While all stages of a typical pipeline are thoroughly described in literature since decades ago, the choice of particular

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algorithms that should be used is still not well defined. Standard linear approaches like PCA (Eigenfaces approach [2]) made it possible to address the problems related to the processing of faces using computationally efficient solutions. Approaches based on statistical measures of texture, like Local Binary Patterns histogram (LBPH [?]), were proposed to provide recognition robust to varying illuminations. Finally, recent developments in deep learning allowed reaching unprecedented performance results using models like FaceNet [4] that learn face representations directly from images.

Despite multiple algorithms being developed to address recognition tasks, researchers and developers still experience significant difficulties in choosing appropriate pipeline for particular use case. The following questions need answers: How accurate face recognition becomes under particular preprocessing approach? What is the scaling efficiency of particular approach? How robust is the model trained on a given face dataset? Answering these questions requires systematic experimentation in consistent settings – which is exactly the focus of our work.

Contributions of the study are listed below:

1. A systematic performance analysis of PCA, LBPH, and FaceNet over three different benchmark datasets, where each dataset is characterized by varying sizes and conditions of acquisition.
2. Comparative performance analysis of six different image processing techniques for each configuration (i.e., model-dataset).
3. Performance analysis metrics in terms of accuracy, precision, recall, F1, PSNR, and SSIM of all 54 different model-dataset-preprocessing technique combinations.
4. Recommendations regarding choice of models and preprocessing technique according to the size of the dataset and the computational budget.

This paper will be structured in the following manner. Literature review is discussed in Section 2. The problem statement is stated in Section 3. Methodology is explained in Section 4. Experimental setup is provided in Section 5. Quantitative results are presented in Section 6. Qualitative analysis is provided in Section 7. Discussion is included in Section 8. Conclusions are drawn

2. Literature Review

2.1. Classical Feature-Based Methods

The Eigenfaces algorithm [2] by Turk and Pentland utilised Principal Component Analysis (PCA) for dimensionality reduction and projection of face images onto a low-dimensional subspace, referred to as face space. It was shown that face images occupy a low-dimensional manifold in the high-dimensional space defined by the pixels and that nearest-neighbour classification in this subspace gives acceptable recognition accuracy in a controlled environment.

Further research considered the application of Linear Discriminant Analysis (LDA)—the Fisherfaces method [5]—to obtain projections that maximise the ratio of between-class scatter to within-class scatter, which yielded better results than PCA if more than one sample per person is used for training.

2.2. Texture and Local Pattern Descriptors

Ojala et al. proposed Local Binary Patterns (LBP) [6], a computationally efficient method for texture description. In [3], Ahonen et al. extended LBP to face description by dividing a face image into several blocks, calculating LBP histograms for each block, and concatenating histograms into a feature vector. The resulting Local Binary Patterns Histograms (LBPH) feature description was resistant to monotonic variations in illumination, unlike global descriptors, and became a standard benchmark in face recognition experiments.

2.3. Deep Learning Approaches

The emergence of large-scale annotated databases of faces, such as LFW [8] and MS-Celeb-1M [?], facilitated training of deep convolutional neural networks to recognise faces with unprecedented accuracy. In their paper, Taigman et al. [7] described DeepFace, a nine-layer neural network combined with 3D face alignment that provided human-like performance on LFW. Schroff et al. [4] from Google proposed FaceNet, a deep network trained using the triplet loss function, which directly optimises the embedding space for minimum intra-class distance and maximum inter-class distance. FaceNet reached 99.63% accuracy on LFW with 128-dimensional embeddings and remains a widely used architecture in practice.

2.4. Image Preprocessing for Face Recognition

Image preprocessing plays a crucial role in the quality of recognition. Yang et al. [1] detail a complete pipeline of image preprocessing that includes transforming an image to grayscale, applying median filtering, performing histogram equalisation, normalising the geometry of an image, and conducting wavelet transforms. Converting an image to grayscale reduces its complexity, and histogram equalisation redistributes the pixel intensities to increase the contrast—especially useful when the illumination of a scene is non-uniform. Wavelet transforms produce multi-resolution decompositions of an image that simultaneously account for

2.5. Comparative Studies

Variants of the problem are discussed in previous works comparing face recognition algorithms over different datasets [9, 10]; however, almost all of these works were done prior to the age of Deep Learning or only used one dataset to compare algorithms. Some work on comparing deep neural network architectures is performed, e.g., in [11], but not much work exists on comparing classic algorithms to deep networks under common preprocessing conditions on multiple heterogeneous benchmark datasets.

3. Problem Statement

For a dataset $\mathcal{D} = \{(I_i, y_i)\}_{i=1}^N$, where $I_i \in \mathbb{R}^{H \times W}$ is a grayscale face image and $y_i \in \{1, \dots, C\}$ is the identity class of the person, face recognition can be seen as an exercise in finding a function $f: \mathbb{R}^{H \times W} \rightarrow \{1, \dots, C\}$ to assign correct identities to images from the dataset.

Formally, we investigate the influence of two components of the face recognition pipeline - preprocessing function $\phi(\cdot)$ and recognition model M on performance of the algorithm:

$$\hat{y} = M(\phi(I)) \quad (1)$$

where $\phi \in \{\text{Original, Median, ...}\}$ and $M \in \{\text{PCA+kNN, LBPH, FaceNet}\}$.

Relevant research questions include:

- RQ1: In what way does accuracy of recognition differ between PCA, LBPH, and FaceNet when applied to the same dataset with the same preprocessing method?
- RQ2: For each recognition model considered, which is the best preprocessing method?
- RQ3: What impact do scale and complexity of the dataset have on model performance?
- RQ4: Is there a trade-off between image quality, as measured in PSNR or SSIM, and recognition accuracy for a preprocessing method?

4. Proposed Methodology

4.1. Image Preprocessing Pipeline

All images are converted to grayscale and resized to 100×100 . Six preprocessing algorithms are applied, including:

- (1) Original: Simple preprocessing consisting of grayscaling and resizing; serves as a reference.
- (2) Median Filtering: Images are processed using a 3×3 median filter. Salt-and-pepper type noise is suppressed, while edges remain untouched [1]. The pixel replacement in the image is expressed as:

$$g(x,y) = \text{median}\{f(x + s,y + t) \mid (s,t) \in W\} \quad (2)$$

where W is a 3×3 neighbourhood window.

- (3) Gray Normalization (Histogram Equalization Variant): Grayscale transformation based on luminosity weights with:

$$\text{gray} = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B \quad (3)$$

followed by normalization via contrast stretching to achieve uniformity of pixel intensity.

- (4) Geometric normalization: Affine transforms—rotation normalization, scaling to standardized size, and centering translations—are performed to align facial features in consistent locations across all input images.

- (5) Wavelet Transform: Multi-resolution image features are extracted using Daubechies wavelet transformation. Level-1 approximation coefficients are kept and reconstructed into an image which contains low-frequency image structure features but excludes noise.

- (6) Histogram Equalization: Histogram equalization is performed through normalization of cumulative density function (CDF) of pixel intensity distribution as follows:

$$s_k = \sum_{j=0}^k (L - 1) \times p_r(r_j) \quad (4)$$

where L is the total number of grey levels, and $p_r(r_j)$ is the normalized histogram. This improves overall contrast of images, especially those that are either under- or over-exposed [1].

4.2. PCA-Based Recognition (Eigenfaces)

Principal component analysis: PCA reduces the dimensionality of images within the training set by calculating eigenvalues of the covariance matrix. Here $\mathbf{X} \in \mathbb{R}^{N \times d}$ is the flattened image

matrix with dimensionality $d = H \times W$. The mean face $\mathbf{x}^-$ is subtracted, and SVD is applied to obtain the top- k principal components $\mathbf{U} \in \mathbb{R}^{d \times k}$ (eigenfaces). Each image is projected as $\mathbf{z} = \mathbf{U}^T(\mathbf{x} - \mathbf{x}^-)$ and recognition is performed using k -Nearest Neighbours (k -NN) in the reduced space. We use $k = 100$ principal components in all experiments.

Algorithm 1 PCA Face Recognition

Require: Training set $\{(\mathbf{x}_i, y_i)\}$, test image $\mathbf{x}^*$, $k = 100$

- 1: Compute mean face $\mathbf{x}^- = \frac{1}{N} \sum_i \mathbf{x}_i$
 - 2: Centre data: $\mathbf{X}^- = \mathbf{X} - \mathbf{1}\mathbf{x}^{-T}$
 - 3: Compute top- k eigenvectors $\mathbf{U}$ via SVD
 - 4: Project training set: $\mathbf{Z}_{\text{train}} = \mathbf{X}\mathbf{U}^-$
 - 5: Project test image: $\mathbf{z}^* = (\mathbf{x}^* - \mathbf{x}^-)^T \mathbf{U}$
 - 6: Assign label via 1-NN: $\hat{y} = y_{\text{argmin}_i \|\mathbf{z}^* - \mathbf{z}_i\|}$
 - 7: return $\hat{y}$
-

4.3. LBPH-Based Recognition

LBPH encodes local texture by comparing each pixel to its P circularly symmetric neighbours at radius R . The binary code for pixel (x_c, y_c) is:

$$\text{LBPPR}(x_c, y_c) = \sum_{p=0}^{P-1} s(gp - gc) \cdot 2^p \quad (5)$$

where $s(u) = 1$ if $u \geq 0$ else 0, g_c is the central pixel value, and g_p are the sampled neighbours. The face image is divided into 8×8 grid regions; the LBP histogram of each region is computed and concatenated. Classification is performed by Chi-squared distance between the probe and gallery histograms. OpenCV's `face.LBPHFaceRecognizer` is used in this work with parameters $P = 8$, $R = 1$, grid size 8×8 .

Algorithm 2 LBPH Face Recognition

Require: Training set $\{(I_i, y_i)\}$, test image I^*

- 1: For each training image I_i , compute $\text{LBP}_{8,1}$ map
 - 2: Divide LBP map into 8×8 grid regions
 - 3: Compute and concatenate regional histograms $\mathbf{h}_i$
 - 4: For test image I^* , compute probe histogram $\mathbf{h}^*$
 - 5: Compute χ^2 distance: $d_i = \chi^2(\mathbf{h}^*, \mathbf{h}_i)$
 - 6: Assign label: $\hat{y} = y_{\text{argmin}_i d_i}$
 - 7: return $\hat{y}$
-

4.4. FaceNet-Based Recognition

FaceNet uses an Inception-ResNet-v1 backbone pre-trained on the VGGFace2 dataset to map face images to a 128-dimensional ℓ_2 -normalised embedding space. The network is trained using the triplet loss:

$$\mathcal{L} = \sum_i [\|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha]_+ \quad (6)$$

where x^a , x^p , x^n are anchor, positive, and negative samples respectively and α is the margin. In our pipeline, the pre-trained model is used as a fixed feature extractor; embeddings are classified using a Support Vector Machine (SVM) with an RBF kernel trained on the embedding space. Images are resized to 160×160 pixels and converted to RGB for FaceNet input.

Algorithm 3 FaceNet Face Recognition

Require: Training set $\{(I_i, y_i)\}$, test image I^* , pre-trained FaceNet $f(\cdot)$

- 1: Resize all images to 160×160 and convert to RGB
 - 2: Extract embeddings: $\mathbf{e}_i = f(I_i) \in \mathbb{R}^{128}$
 - 3: Train SVM classifier on $\{(\mathbf{e}_i, y_i)\}$
 - 4: Extract probe embedding: $\mathbf{e}^* = f(I^*)$
 - 5: Predict: $\hat{y} = \text{SVM}(\mathbf{e}^*)$
 - 6: return $\hat{y}$
-

4.5. System Architecture

The overall system workflow is described as follows. Raw images are loaded from the dataset directory, converted to grayscale, resized to a standard dimension, and split into training (80%) and test (20%) sets using stratified random sampling. Each of the six preprocessing functions is applied to both splits. The processed images are then forwarded to each recognition model for training and inference. Performance metrics are computed on the held-out test set and aggregated across preprocessing methods.

5. Experimental Setup

5.1. Datasets

AT&T Database of Faces (ORL): Originally compiled at AT&T Laboratories Cambridge, this dataset contains 400 grayscale images of 40 distinct subjects (10 images per subject) at 92×112 pixels. Images were captured under varying lighting, facial expressions (open/closed eyes, smiling/non-smiling), and facial details (glasses/no glasses). The dataset is a widely used benchmark for controlled-setting face recognition evaluation.

FEI Face Dataset: A Brazilian face database containing 2,800 colour images of 200 subjects (14 images per subject), captured at the FEI Institute in São Bernardo do Campo, Brazil. Images include frontal and profile views, with variations in expression (neutral and smiling), illumination, and background. Images were resized to 100×100 pixels in grayscale for this study.

Faces94 Dataset: Collected at the University of Essex, this dataset contains 3,059 images of 152 subjects with approximately 20 images per subject. Images are colour JPEG files captured in a constrained office environment with variation in expression, slight head movement, and minor lighting changes. Images were resized to 100×100 pixels in grayscale.

Table 1: Dataset Characteristics

Dataset	Subjects	Images	Imgs/Subject	Resolution
AT&T (ORL)	40	400	10	92 × 112
FEI Face	200	2,800	14	640 × 480
Faces94	152	3,059	~20	180 × 200

5.2. Train-Test Split

An 80/20 stratified train-test split was applied uniformly across all datasets to ensure each subject appears in both partitions. Stratification guarantees that class proportions in the test set mirror those in the full dataset.

- AT&T: 320 training / 80 test images
- FEI: 2,240 training / 560 test images
- Faces94: 2,447 training / 612 test images

5.3. Software and Hardware

Experiments were conducted in Google Colaboratory using Python 3.12. Key libraries include OpenCV 4.13 (cv2), scikit-learn 1.6.1, scikit-image 0.25.2, PyWavelets 1.9.0, Keras-FaceNet 0.3.2 (wrapping TensorFlow 2.x), pandas 2.2.2, and matplotlib 3.10.0. Hardware: NVIDIA T4 GPU (16 GB VRAM), 12.7 GB RAM.

5.4. Evaluation Metrics

All metrics are computed on the held-out test set:

- Accuracy: $\frac{TP+TN}{Total}$
- Precision $\frac{1}{C} \sum_c \frac{TP_c}{TP_c + FP_c}$ (macro):
- Recall $\frac{1}{C} \sum_c \frac{TP_c}{TP_c + FN_c}$ (macro):
- F1 Score (macro): Harmonic mean of precision and recall
- PSNR: $10 \log_{10} \frac{MAX^2}{MSE}$ — measures image fidelity after preprocessing
- SSIM: Structural similarity between original and preprocessed images

6. Quantitative Analysis

6.1. FaceNet Results

Table 2: FaceNet Performance on AT&T Dataset (%)

Method	Acc.	Prec.	Recall	F1
Original	100.00	100.00	100.00	100.00
Median	100.00	100.00	100.00	100.00
GrayNorm	100.00	100.00	100.00	100.00
Geometric	100.00	100.00	100.00	100.00
Wavelet	71.50	77.82	71.50	70.21
Histogram	100.00	100.00	100.00	100.00

Table 3: FaceNet Performance on FEI Dataset (%)

Method	Acc.	Prec.	Recall	F1
Original	88.43	94.56	88.43	90.45
Median	84.07	92.43	84.07	86.60
GrayNorm	91.50	94.38	91.50	92.21
Geometric	87.57	94.24	87.57	89.67
Wavelet	55.14	64.20	55.14	56.64
Histogram	86.64	92.30	86.64	88.21

Table 4: FaceNet Performance on Faces94 Dataset (%)

Method	Acc.	Prec.	Recall	F1
Original	99.35	98.75	99.35	99.04
Median	99.35	98.75	99.35	99.04
GrayNorm	99.35	98.75	99.35	99.04
Geometric	99.35	98.75	99.35	99.04
Wavelet	97.45	97.15	97.45	97.14
Histogram	99.35	98.75	99.35	99.04

6.2. LBPH Results

Table 5: LBPH Performance on AT&T Dataset (%)

Method	Acc.	Prec.	Recall	F1
Original	95.83	97.07	97.99	96.79
Median	95.83	96.43	97.99	96.53
GrayNorm	95.83	97.07	97.99	96.79
Geometric	95.83	96.43	97.99	96.74
Wavelet	93.33	92.83	93.72	92.59
Histogram	95.83	97.07	97.99	96.79
Gaussian	95.83	97.07	97.99	96.79

Table 6: LBPH Performance on FEI Dataset (%)

Method	Acc.	Prec.	Recall	F1
Original	88.21	88.32	88.65	87.14
Median	78.33	80.35	78.53	76.52
GrayNorm	87.74	87.68	87.86	86.17
Geometric	88.33	87.53	88.13	86.58
Wavelet	90.00	91.17	91.38	90.03
Histogram	87.50	88.71	88.14	86.83
Gaussian	83.69	84.79	84.21	82.47

Table 7: LBPH Performance on Faces94 Dataset (%)

Method	Acc.	Prec.	Recall	F1
Original	98.37	98.62	98.24	98.31
Median	98.26	98.62	98.17	98.26
GrayNorm	98.04	98.62	97.98	98.14
Geometric	98.26	98.62	98.15	98.25
Wavelet	98.37	98.62	98.24	98.31

Histogram	98.47	98.62	98.33	98.36
Gaussian	98.26	98.62	98.17	98.26

6.3. PCA Results

Table 8: PCA Performance on AT&T Dataset (%)

Method	Acc.	Prec.	Recall	F1
Original	96.00	96.90	96.00	95.93
Median	96.00	96.90	96.00	95.93
GrayNorm	97.00	97.41	97.00	96.96
Geometric	95.50	96.72	95.50	95.45
Wavelet	96.00	96.90	96.00	95.93
Histogram	95.00	96.18	95.00	95.03

Table 9: PCA Performance on FEI Dataset (%)

Method	Acc.	Prec.	Recall	F1
Original	80.36	82.88	80.36	80.09
Median	80.36	83.13	80.36	80.04
GrayNorm	87.86	90.45	87.86	87.69
Geometric	80.64	83.36	80.64	80.44
Wavelet	81.64	84.24	81.64	81.48
Histogram	87.29	88.87	87.29	87.00

Table 10: PCA Performance on Faces94 Dataset (%)

Method	Acc.	Prec.	Recall	F1
Original	98.37	98.63	98.37	98.33
Median	98.37	98.63	98.37	98.33
GrayNorm	98.43	98.63	98.43	98.44
Geometric	98.30	98.63	98.30	98.30
Wavelet	98.43	98.69	98.43	98.40
Histogram	98.30	98.63	98.30	98.29

6.4. Cross-Model Summary

Table 11 illustrates the highest classification accuracy obtained for each preprocessing method on each dataset for each of the models considered.

Table 11: Best Accuracy (%) per Model per Dataset

Dataset	FaceNet	LBPH	PCA
AT&T (ORL)	100.00	95.83	97.00
FEI Face	91.50	90.00	87.86
Faces94	99.35	98.47	98.43

6.5. Image Quality Metrics

Table 12 illustrates PSNR and SSIM values averaged for the representative preprocessing methods, which have been selected based on their performance. Median filter gives the highest PSNR value (≈ 34.7 dB), meaning the least distortions of the pixel intensities compared to the original image. Gray normalization shows the highest SSIM value on Faces94 (≈ 0.993), meaning the highest structural similarity to the original image. Finally, wavelet transform gives

the lowest PSNR and SSIM values, which is in line with its strong frequency decomposition properties.

Table 12: Mean PSNR and SSIM by Preprocessing Method (Faces94)

Preprocessing Method	PSNR (dB)	SSIM
Original	∞	1.000
Median	33.98	0.963
GrayNorm	34.54	0.993
Geometric	18.48	0.656
Wavelet	9.91	0.439
Histogram	15.15	0.723

7. Qualitative Analysis

7.1. Effect of Preprocessing on Recognition

The qualitative evaluation of the preprocessing outputs, which agrees with the findings presented in [1], indicates that different preprocessing methods result in different output qualities. For example, median filter smoothens out the speckle noise but does not affect the edge sharpness, unlike the mean filter. Gray normalization adjusts the intensity distribution in the histogram of an image to achieve more uniform perception—an important property for applications under different illumination conditions. Geometrical normalization aligns facial landmarks, such as the eyes, nose, and mouth, making the intraclass variance minimal in terms of the head pose.

Finally, wavelet transform results in highly distorted images with low SSIM values around 0.33–0.55. This low quality of the transformed images explains the degradation of the face recognition accuracy, especially in the case of FaceNet, since the embeddings produced by this model rely on fine-grained features.

7.2. Dataset-Level Observations

AT&T is the smallest of the three datasets, consisting of 400 images, with 40 subjects only. Given the controlled acquisition conditions, this small dataset allows all three models to achieve high accuracy with no preprocessing.

The FEI dataset presents a much more difficult test case: with 200 subjects and 14 images per subject, the presence of both expression variations (neutral and smiling) and profile poses introduces significant inter-pose and inter-expression variances, which negatively impact the PCA model.

The Faces94 dataset has the largest collection of subjects (152 subjects with ~ 20 images per subject). It performs well for all three models under all preprocessing techniques. The reason behind this high performance might be the relatively large number of samples per subject after splitting into the training set (about 16 images).

7.3. Qualitative Comparison with Base Paper

The pioneering work by Yang and Chen [1] highlights that face recognition outcomes are highly dependent on the composition of an image: images featuring a centered portrait produce significantly better recognition results compared to off-centered or badly-framed faces. According to their qualitative evaluation using MATLAB, grayscale processing combined with minimization of background through candidate box selection performs well for center-

aligned faces. These findings are confirmed by our quantitative tests—the contrast enhancing and background minimizing methods (gray normalization, histogram equalization) consistently outperformed methods distorting image information (wavelet transform, geometric normalization in extreme transformations).

Furthermore, one of the drawbacks identified in the pioneering study was background colors and intensities matching skin tone color and intensity, resulting in poor recognition performance—an issue effectively addressed by histogram equalization and gray normalization—the two most successful preprocessing techniques used in our testing.

8. Results and Discussion

8.1. FaceNet vs. LBPH vs. PCA

FaceNet attains the highest face recognition accuracies (AT and T: 100%, Faces94: 99.35%) thanks to its deep Inception-ResNet architecture pre-trained on millions of faces. 128-dimensional embedding space learned by the model enables discrimination of highly discriminative features resistant to changes of lighting and facial expressions within the training distribution.

LBPH shows significantly superior performance on the FEI database (90.00% accuracy) after waveletbased preprocessing and surpasses FaceNet by a large margin (55.14%). This is explained by the property of LBPH to be illumination-invariant as local binary coding learns the relationships between pixels in a patch of the image, making it robust to any distortions in frequencies introduced by wavelet decomposition. Additionally, LBPH proves to have the lowest variance among all models on the Faces94 dataset after any kind of preprocessing (98.04% - 98.47%).

PCA is the most computationally efficient method and the least accurate in recognizing faces on the complex FEI (80.36%) dataset suffering from intra-class variability—a known problem of PCA. However, when provided with sufficient samples, such as in the controlled Faces94 dataset, PCA competes with LBPH and is very close to FaceNet (around 98.4%).

8.2. Best Preprocessing Strategy

Gray normalization (histogram equalized grayscale image) proved to be the best preprocessing technique along with histogram equalization.

- PCA on AT&T: GrayNorm results in 97.00
- FaceNet on FEI: GrayNorm results in 91.50
- PCA on FEI: GrayNorm results in 87.86

The most ineffective pre-processing approach appears to be wavelet transforms, with accuracy reductions reaching up to 33 percentage points as compared to the baseline results. Namely, on FaceNet on FEI, it yields an accuracy of 55.14

8.3. Dataset Complexity and Scale

It becomes apparent that all three approaches reach their maximum performance on Faces94 (largest database of the three), demonstrate relatively high performance on AT&T, and have the largest difference in results obtained for the FEI dataset. Indeed, FEI, with its presence of profile pictures, highly varying expression and large number of subjects (200),

represents the most difficult recognition task. This clearly highlights the importance of using diverse databases for the benchmarks and shows that recognition performance on a limited database may vary greatly from the one observed in real-life scenarios.

8.4. Practical Recommendations

Given the experimental results, we suggest:

- In case of resource-constrained deployment (edge devices): use PCA with gray normalization – gets 97
- On a server where high precision is required: use FaceNet with gray normalization;
- High-accuracy server-side applications: Use FaceNet with gray normalization.
- Avoid wavelet pre-processing for FaceNet for databases with a high number of subjects.

9. Conclusion

In this paper, we presented a thorough comparative analysis of three face recognition models: PCA, LBPH, and FaceNet. We considered the effect of six different pre-processing approaches on the recognition results when applying the aforementioned models on three benchmark databases (AT&T, FEI, Faces94). After performing 54 controlled experiments and evaluating the resulting precision, recall and F-score, the following conclusions can be made:

First, FaceNet outperforms other approaches on the controlled databases, achieving 100

Notably, the selection of preprocessing steps can significantly affect the accuracy of facial recognition systems. While gray normalization and histogram equalization always improve the accuracy of the networks, wavelet transform significantly reduces the performance level, especially in FaceNet for FEI, whose accuracy dropped from 88.43

10. Future Work

A number of research directions remain open in this regard. First, expand the experiment to incorporate the use of advanced structures like ArcFace [deng2019arcface] and AdaFace [kim2022adaface], which utilize advanced margin losses. Second, study the effects of data augmentation (random cropping, flipping, color jittering) on the accuracy of facial recognition systems, especially with small databases such as ATT. Third, test different approaches to fine-tuning the FaceNet model on the target datasets to minimize the discrepancy between the training dataset from VGGFace2 and the target sets. Fourth, investigate real-world constraints in implementing a facial recognition algorithm, including inference speed, memory overheads, and accuracy in quantized and pruned models.

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